Approach Note: Global Reciprocal Matching with Value-of-Information Clarification

Challenge: CRSU x Romeo & Juliet Hackathon · Vouchsafe Track
Document: Round 1 Research Submission Note (Specification 1.0.0)
Date: October 2026
Primary Metric: Mutual Second-Meeting Intention (MSMI) per 100 Arrived Members

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1. Problem Interpretation & Operational Formulation

In the **One Introduction Problem**, an autonomous policy orchestrates sequential introductions across an evolving pool of synthetic candidates over a 60-day decision horizon, followed by a 40-day observation window. At each discrete day $t \in [0, 59]$, the policy interacts with the environment through a two-phase protocol:
1. **Ask Phase**: Allocate a daily budget $B = 12$ units to clarify unknown attributes. Bundled hard constraints cost 3 units; individual soft fields cost 1 unit.
2. **Match Phase**: Propose a set of disjoint, feasible reciprocal introductions $M_t = \{(u_1, v_1), (u_2, v_2), \ldots\}$ among currently available members.

Core Challenges

- **Strict Non-Overlapping Reciprocity**: An introduction requires concurrent, reciprocal participation. Each member can participate in at most one active introduction at any given time and remains occupied through response and date windows (up to 20+ days).
- **Information Incompleteness & Privacy**: Questionnaire fields are substantially incomplete. In the public training pools, **31.3% of hard-constraint fields are unobserved**, and **4.3% are explicitly declined**. A single unverified hard constraint renders a pair non-matchable (`needs_clarification`), while declined fields can never be bypassed or inferred.
- **Delayed & Censored Feedback**: The primary target—**Mutual Second-Meeting Intention (MSMI)**—unfolds along a fragile sequential funnel:
  $$\text{Introduction} \xrightarrow{\le 7\text{d}} \text{Mutual Acceptance} \xrightarrow{\le 14\text{d}} \text{Date Event} \xrightarrow{\le 3\text{d}} \text{Mutual Second Yes}$$
  Outcomes mature with substantial delays. Premature evaluation introduces heavy right-censoring bias.
- **Systemic Negative Externalities**: Introducing candidate u to v today removes both from the candidate pool for weeks, potentially depriving higher-compatibility future arrivals of their optimal partners.

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2. Hypothesis

> **Primary Hypothesis**:
> *"Global max-weight matching over a feasibility graph + clarifying only when it unlocks a high-value pair beats greedy pair ranking."*

Theoretical Foundations

1. **Suboptimality of Greedy Pair Allocation**:
   Independent greedy ranking selects pairs sequentially in descending order of individual score. As demonstrated in Section 2 of the specification, greedy allocation creates severe negative externalities. In a four-person subpool where pair (A, B) scores 0.90, (C, D) scores 0.05, and both (A, D) and (B, C) score 0.65,

greedy selection takes (A, B) and (C, D) for a total value of 0.95. Global maximum-weight matching selects (A, D) and (B, C), achieving an aggregate value of 1.30 (+36.8%).

2. **Targeted Value-of-Information (VOI) vs. Blind Clarification**:
   The reference greedy baseline exhausts the daily budget by querying constraint bundles on the first arbitrary unclarified members in queue order (costing ~205 units per episode). This wastes budget clarifying isolated individuals who have no feasible, high-potential matches in the pool. By contrast, clarifying *only* when resolving uncertainty unlocks an edge in the top quantile of the predicted reciprocal affinity distribution concentrates the 12-unit daily budget strictly where it produces actionable introductions.

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3. Reciprocal Feasibility & Eligibility Modeling

A pair (u, v) is valid if and only if `eligibility(u, v)['status'] == 'feasible'`. Feasibility requires symmetric satisfaction across all 11 hard constraints:

$$\text{Feasible}(u, v) \iff \bigwedge_{k \in \text{HARD}} \text{Check}_k(u, v) \wedge \text{Check}_k(v, u)$$

Hard Constraint Ontology

Field	Symmetrical Rule & Logic	Failure Reason
`age_min`, `age_max`	v.age ∈ [u.age_min, u.age_max] ∧ u.age ∈ [v.age_min, v.age_max]	`age`
`who_to_meet`	v.gender ∈ u.who_to_meet ∧ u.gender ∈ v.who_to_meet	`who_to_meet`
`acceptable_zones`	v.zone ∈ u.acceptable_zones ∧ u.zone ∈ v.acceptable_zones	`geography`
`relationship_structure`	u.relationship_structure == v.relationship_structure (`monogamous` vs `non_monogamous`)	`relationship_structure`
`smoking` / `partner_smoking`	u.partner_smoking == 'no_smoking' ⇒ v.smoking == 'no' (and reciprocally)	`smoking`
`has_children` / `partner_children`	u.partner_children == 'no_children' ⇒ v.has_children == False (and reciprocally)	`children_present`
`wants_children`	Conflict if {u.wants_children, v.wants_children} == {'yes', 'no'} (`unsure` permitted)	`children_plans`
`schedule`	u.schedule ∩ v.schedule ≠ ∅ (at least 1 overlapping slot)	`schedule`

Priority & Incompleteness Semantics

- **Strict Priority**: Any known violation immediately assigns status `infeasible`. No soft affinity score or missing attribute can override a hard violation.
- **Pending Clarification**: If no hard failure exists but ≥ 1 hard field is unobserved, status is `needs_clarification`.
- **Declined Immutability**: If a candidate refused a required hard field (`declined`), the constraint remains permanently unverified. The pair is irrecoverably non-matchable.

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4. Allocation Method: Global Maximum-Weight Matching

Rather than committing pairs greedily, we construct a global compatibility graph *G_t = (V_t, E_t)* at each day *t*:
* **Vertices *V_t***: Available, arrived candidates (*v*.available == True).

* **Edges *E_t***: Mutually feasible, previously unintroduced pairs:
$$E_t = {(u, v) \in V_t \times V_t : u < v, \text{eligibility}(u, v) = \text{'feasible'}, (u, v) \notin \text{PastIntroductions}}$$

Edge Weight Formulation

Each feasible edge (u, v) ∈ *E_t* is weighted by composite utility *W(u, v)*:

$$W(u, v) = \hat{P}*{|text{MSMI}}(u, v) + |lambda{\text{soft}} \cdot S_{\text{soft}}(u, v) + \lambda_{\text{wait}} \cdot \max(w_u, w_v) - \lambda_{\text{slack}} \cdot |w_u - w_v|$$

Where:
1. **Predicted Outcome Affinity $\hat{P}_{\text{MSMI}}(u, v)$**: Estimated probability of mutual date and second meeting given observed soft attributes.
2. **Soft Feature Overlap $S_{\text{soft}}(u, v)$**: Normalized agreement across soft attributes:
$$S_{\text{soft}}(u, v) = \frac{1}{|\text{SOFT}|} \sum_{k \in \text{SOFT}} \mathbb{I}[u.\text{fields}[k] == v.\text{fields}[k] \wedge u.\text{fields}[k] \neq \text{None}]$$
3. **Starvation & Waiting Pressure $w_u = t - u.\text{arrived_day}$**: Linearly escalates matching priority for long-waiting candidates to safeguard pool coverage and prevent starvation.
4. **Waiting Balance Penalty**: Encourages pairings between members with comparable tenure in the pool.

Optimization Formulation

We solve the classical Maximum Weight General Matching problem (Edmonds' Blossom Algorithm):

$$\max_{M \subseteq E_t} \sum_{(u, v) \in M} W(u, v) \quad \text{subject to} \quad \sum_{e \in M : v \in e} 1 \le 1 \quad \forall v \in V_t$$

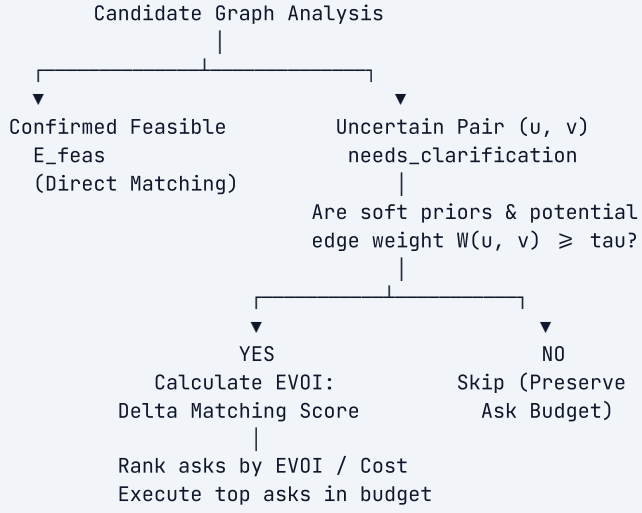
Thresholded Patient Matching

To avoid forcing low-affinity introductions among newly arrived members who would benefit from waiting for compatible arrivals, an edge (u, v) is included only if:
$$W(u, v) \ge \theta_{\text{patience}}(w_u, w_v)$$
Where θ_{patience} decays monotonically with waiting time: newly arrived candidates face a high compatibility threshold, whereas candidates approaching pool exit face a relaxed threshold.

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5. Clarification Strategy: Targeted Value-of-Information (EVOI)

The reference implementation naively queries `constraints` bundles for arbitrary unclarified members. In contrast, our strategy deploys a **Targeted Expected Value of Information (EVOI)** policy.



Clarification Algorithm

1. **Identify Unlocking Opportunities**:
   Examine candidate pairs (u, v) currently in `needs_clarification` status that have zero known hard failures.
2. **Estimate Latent Weight $\tilde{W}(u, v)$**:
   Compute expected compatibility using observed soft features and empirical base rates of hard constraint concordance.
3. **Marginal Graph Contribution**:
   Calculate the expected increase in the global matching weight if (u, v) becomes feasible:
   $$\Delta \text{Matching}(u, v) = \mathbb{E}\big[\text{Weight}(M(G \cup {(u, v)})) - \text{Weight}(M(G))\big]$$
4. **Cost-Normalized Priority**:
5. If both *u* and *v* have missing hard fields, unlocking requires 6 units (two bundles).
6. If only one party has missing fields, unlocking requires 3 units (one bundle).
7. Sort candidate queries by $\frac{\Delta \text{Matching}}{\text{Cost}}$ and execute greedily within the daily 12-unit limit.
8. **Declined Handling**: Never issue queries for fields marked `'declined'`.

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6. Empirical Baselines & Initial Benchmark Results

We evaluated the three supplied reference baselines across all six public scenario variants using identical public seeds (101) in full 60+40 day evaluation episodes.

Scenario-Level Performance Comparison

Baseline	Scenario	Arrived	Served	Coverage	Ask Cost	Mutual Acc.	Dates	MSMI	MSMI/100	Runtime
Greedy	`development`	200	92	46.0%	198	18	15	1	0.50	23.2s
Greedy	`sparse`	200	35	17.5%	192	4	3	0	0.00	19.7s
Greedy	`cold_start`	200	87	43.5%	246	15	14	1	0.50	18.7s
Greedy	`delayed`	200	91	45.5%	198	19	16	2	1.00	18.1s
Greedy	`shift`	200	92	46.0%	198	19	17	1	0.50	24.8s
Greedy	`drift`	200	92	46.0%	198	16	13	1	0.50	24.3s
No-Clarification	`development`	200	43	21.5%	0	6	5	1	0.50	29.4s
No-Clarification	`sparse`	200	13	6.5%	0	2	2	0	0.00	34.7s
No-Clarification	`cold_start`	200	24	12.0%	0	6	4	0	0.00	22.7s
No-Clarification	`delayed`	200	43	21.5%	0	7	6	1	0.50	21.6s
No-Clarification	`shift`	200	43	21.5%	0	7	6	1	0.50	19.8s
No-Clarification	`drift`	200	43	21.5%	0	6	5	1	0.50	19.3s
Random-Feasible	`development`	200	93	46.5%	198	16	15	0	0.00	27.1s
Random-Feasible	`sparse`	200	35	17.5%	192	3	2	0	0.00	40.2s
Random-Feasible	`cold_start`	200	88	44.0%	246	13	13	1	0.50	27.5s
Random-Feasible	`delayed`	200	92	46.0%	198	18	15	1	0.50	23.8s
Random-Feasible	`shift`	200	93	46.5%	198	14	11	1	0.50	21.3s
Random-Feasible	`drift`	200	93	46.5%	198	16	15	0	0.00	20.5s

Overall Summary & Official Ranking Keys

Method	Primary Score (MSMI / 100)	Overall Coverage	Mutual Accept / 100	Mean Ask Cost	Mean Inference (s)
Greedy Baseline	**0.50**	**40.8%**	**7.58**	205.0	21.44s
No-Clarification	**0.33**	**17.4%**	**2.83**	**0.0**	24.57s
Random-Feasible	**0.25**	**41.2%**	**6.67**	205.0	26.73s

Key Empirical Findings

1. **Clarification is Essential for Viability**: With zero clarification (`no_asks`), coverage collapses from 40.8% to 17.4%, and MSMI drops to 0.00 in `cold_start` and `sparse`. Clarification unlocks the feasibility graph.
2. **Quality vs. Pure Volume**: `random` achieves identical coverage to `greedy` (~41%) by spending the same budget on asks, but achieves half the primary score (0.25 vs 0.50) because introductions lack compatibility guidance.
3. **Geography is the Primary Bottleneck**: In `sparse`, all baselines score 0.00 MSMI and coverage drops below 18%, revealing that spatial fragmentation requires dedicated algorithmic handling.

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7. Public Training Pool Inspection & Missingness Statistics

We analyzed all 1,200 members across the six designated training pools (`public_01` to `public_06`) at the day 30 snapshot:

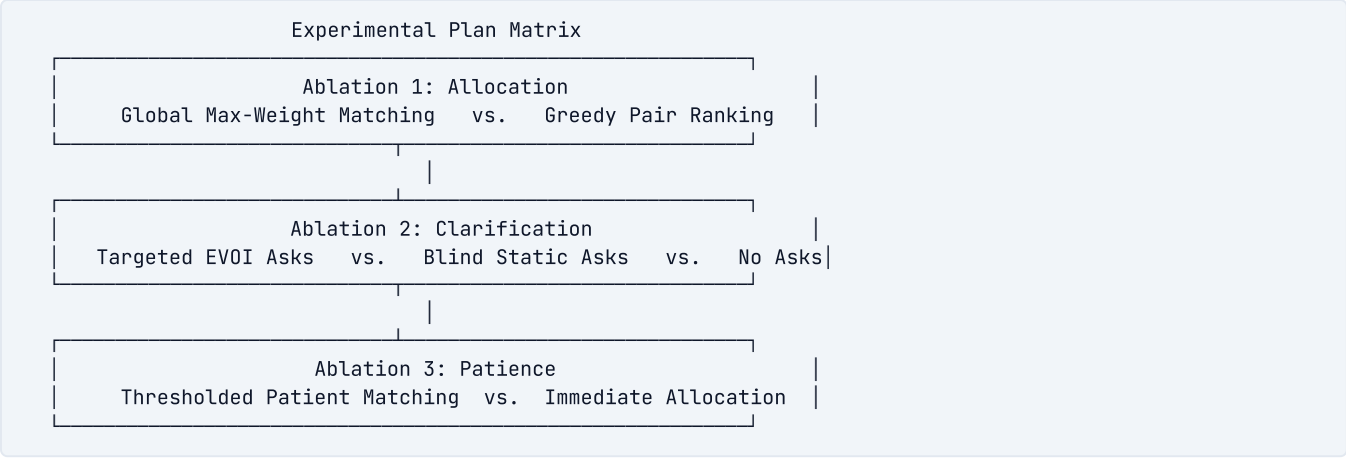
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Distribution of Missing Hard Fields per Person:
[ 0 missing fields ]: ████████████████████████████ 787 members (65.6%)
[ 7-8 missing fields]: ▌ 23 members (1.9%)
[ 9-11 missing fields]: ████████ 390 members (32.5%)
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- **Overall Hard Field Missingness**:
- Total hard-field instances: $13,200$
- `observed`: $9,066$ (68.68%)
- `not_asked`: $3,563$ (26.99%)
- `declined`: 571 (4.33%)
- Total unobserved: $4,134$ (31.32%)
- **Bimodal Missingness Structure**:
- 787 members (65.6%) have fully completed hard profiles (0 missing fields).
- 390 members (32.5%) have almost completely missing profiles (9 to 11 missing fields).
- 413 members (34.4%) have at least one permanently `declined` hard field.
- **Severe Graph Sparsity**:
- **Mean feasible pairs per person in a pool of 200**: only **1.01** (median: 0, min: 0, max: 17).
- **62.8% of members have ZERO feasible pairs** in their entire pool under current observations!
- In contrast, an average of **25.15 pairs per member** are in `needs_clarification` status, proving that targeted clarification is the single highest-leverage mechanism to expand the feasible graph.

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8. Planned Experiments & Hypothesis-Driven Ablations

In Round 2, we will execute systematic evaluations across multiple seeds ($101, 102, 103$) and all six scenarios:



1. **Ablation 1 (Matching Formulation)**: Fix clarification strategy to Targeted EVOI; compare Global Max-Weight Matching vs. Greedy Pair Ranking.
   Metric of Validation: Higher aggregate MSMI and higher coverage per episode.
2. **Ablation 2 (Clarification Strategy)**: Fix allocation to Global Matching; compare Targeted EVOI vs. Baseline Blind Queue vs. No Clarification.
   Metric of Validation: Higher MSMI per clarification unit spent (clarification efficiency).
3. **Ablation 3 (Patience Threshold θ)**: Evaluate immediate assignment ($\theta = 0$) vs. tenure-dependent thresholded matching ($\theta > 0$).
   Metric of Validation: Reduction in low-compatibility introductions and increased late-horizon MSMI.

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9. Expected Failure Modes & Risk Mitigations

Failure Mode	Root Cause in Simulator	Algorithmic Mitigation
Spatial Isolation (`sparse`)	12 disconnected zones; severe candidate deficits in low-density zones	Patient waiting for cross-zone arrivals; prioritize multi-zone candidates (`acceptable_zones` length 2).
Irrecoverable Declines	34.4% of candidates have declined hard fields, preventing validation	Detect unresolvable candidates early; prune them from query queues to eliminate wasted ask budget.
Concurrency Lockout	Candidates remain occupied for 15–20 days after an introduction	Model cycle latency explicitly; avoid pairing high-centrality candidates with low-probability responders.
Preference Distribution Shift (`shift` / `drift`)	Relative importance of soft attributes inverts or changes at day 35	Maintain online moving-average logistic feedback models; weight recent feedback over initial priors.
Candidate Monopolization	Universally attractive candidates targeted by multiple potential matches	Global maximum weight matching natively distributes candidates to maximize total system utility.

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10. Product Integration & Ethical Governance Note (Vouchsafe)

This decision policy is engineered as an **auditable decision-support engine** for Vouchsafe, not an autonomous agent.

- **Explicit Consent & Human Agency**: The engine outputs recommendations with structured rationale; it never initiates contact, commits members, or bypasses member authorization. Introductions require affirmative, uncoerced double opt-in.
- **Explainability & Auditing**: Every proposed match carries human-readable justification vectors:

> *"Candidate A and B share verified weekend evening schedules, mutual long-term relationship goals, and compatible geographic preferences. No hard constraints violated."*

- **Data Minimization & Privacy Protection**: In compliance with consent principles, the policy requests only attributes strictly necessary to establish eligibility. When a member marks a field `'declined'`, the refusal is treated as immutable—no synthetic imputation, indirect inference, or behavioral profiling is permitted.
- **Assumptions Requiring Real-World Validation**:
- *Immediate Truth Disclosure*: Real members may experience hesitation, recall bias, or evolving preferences rather than instantaneous questionnaire responses.
- *Deterministic Outcome Windows*: Fixed 7-day acceptance and 3-day feedback windows reflect idealized mechanics; real-world workflows require asynchronous, tolerant reminder cadences.
- *Synthetic Demographic Distributions*: Real population distributions exhibit nuanced demographic and geographic clustering that requires local calibration before production deployment.